

Journal of the Geological Society of Australia

ISSN: 0016-7614 (Print) Journal homepage: www.tandfonline.com/journals/taje19

Palaeomagnetic results from Mounts Gambier and Schank, South Australia

Mike Barbetti & M. J. Sheard

To cite this article: Mike Barbetti & M. J. Sheard (1981) Palaeomagnetic results from Mounts Gambier and Schank, South Australia, Journal of the Geological Society of Australia, 28:3-4, 385-394, DOI: 10.1080/00167618108729177

To link to this article: https://doi.org/10.1080/00167618108729177

Published online: 01 Aug 2007.

Submit your article to this journal

Article views: 142

View related articles

Citing articles: 3 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=taje20

Journal of the Geological Society of Australia (1981), 28, 385-394
© Geological Society of Australia, Inc., 1981

Palaeomagnetic results from Mounts Gambier and Schank, South Australia

Mike Barbetti[1] & M. J. Sheard[2]

[1] *Radiocarbon Laboratory, Department of Physical Chemistry, University of Sydney, Sydney N.S.W. 2006 (formerly Physics Department, University of Adelaide).*
[2] *Geological Survey, South Australian Department of Mines and Energy, P.O. Box 151, Eastwood S.A. 5063.*

ABSTRACT

Palaeomagnetic results are presented for two young volcanoes in southeastern Australia. Directions for the basalt and lava spatter, erupted near the beginning and end of activity at Mt Gambier respectively, are identical and suggest there was no long break in activity. The direction for basalt at Mt Schank differs from that of Mt Gambier by several degrees, and indicates that the two volcanoes differ in age, probably by three centuries or more. Directions for both volcanoes are much shallower than the present-day geomagnetic direction at the locality. Palaeomagnetic field strength estimates, from baked ash at Mt Gambier and baked sand at Mt Schank, are higher than the present-day field strength. Comparisons with other late Pleistocene and Holocene results suggest that ages for Mts Gambier and Schank are either greater than 7000 years or between 5000 and 1000 years. These conclusions are consistent with published radiocarbon ages from charcoal in fossil soil horizons sealed beneath ash deposits.

INTRODUCTION

Much of central and western Victoria is covered by the Newer Volcanics of late Tertiary to Quaternary age. The edge of the volcanic field extends into the SE corner of South Australia (Fig. 1a), where there are 17 eruptive centres in two distinct groups. Volcanic ejecta from the northern group, a few tens of kilometres to the NW of Mt Gambier, rest on the Gambier Limestone and are overlain by dune sands of the Bridgewater Formation. Solomon (1951) and Sprigg (1952) considered these volcanics to be early to late Pleistocene in age. Volcanics from Mts Gambier and Schank, which comprise the second group (Fig. 1b), rest on the dune sands and are therefore stratigraphically younger than the northern group. The terrain surrounding and underlying the volcanics is flat, with little or no surface drainage and abundant underground water.

Three radiocarbon ages have been published for Mts Gambier and Schank; locations are shown in Figure 1b. Charcoal from the top part of a fossil soil horizon, immediately beneath 1.1 m of ash deposits from Mt Gambier, gave an age of 4830 ± 70 yr b.p. (NZ-33; Fergusson & Rafter, 1957). Another charcoal sample found in sand underneath 30 cm of ash gave an age of 1410 ± 90 yr b.p. (Gak-609; Kigoshi & Kobayashi, 1966). There is nothing to suggest that more than one ash sequence is involved, so the large difference between the dates is puzzling. Possibly the charcoal with the older date may have been buried in the soil for a long period (and hence may pre-date the ash fall), or the charcoal with the younger date may be intrusive (e.g. from a tree root that post-dates the eruption). The age of Mt Gambier should be clarified by further radiocarbon dating now in progress (G. Blackburn, pers. comm.).

The age of Mt Schank is limited by a single radiocarbon date of 18 100 ± 350 yr b.p. on soft charcoal fragments from a soil buried under tuff (ANU-646; Polach et al., 1978). Details of the depth of burial and tuff thickness were not published, but the description appears to rule out the likelihood that the charcoal is intrusive. However, the charcoal could pre-date the volcanic eruption by up to a few thousand years. The date of eruption is being determined by thermoluminescence measurements on baked sand under the lava flow (B. Smith & J. R. Prescott, pers. comm.).

Basalts are widely recognised as being particularly good recorders of the Earth's magnetic field, because they acquire a very stable magnetism as they cool below 580°C. The present study is part of a systematic study of geomagnetic field variations in Australia for the last 50 000 years. The results we have obtained from Mts Gambier and Schank, when compared with other palaeomagnetic results, place new constraints on their possible ages.

MS. received 16 October 1980; revised MS. received 3 April 1981.

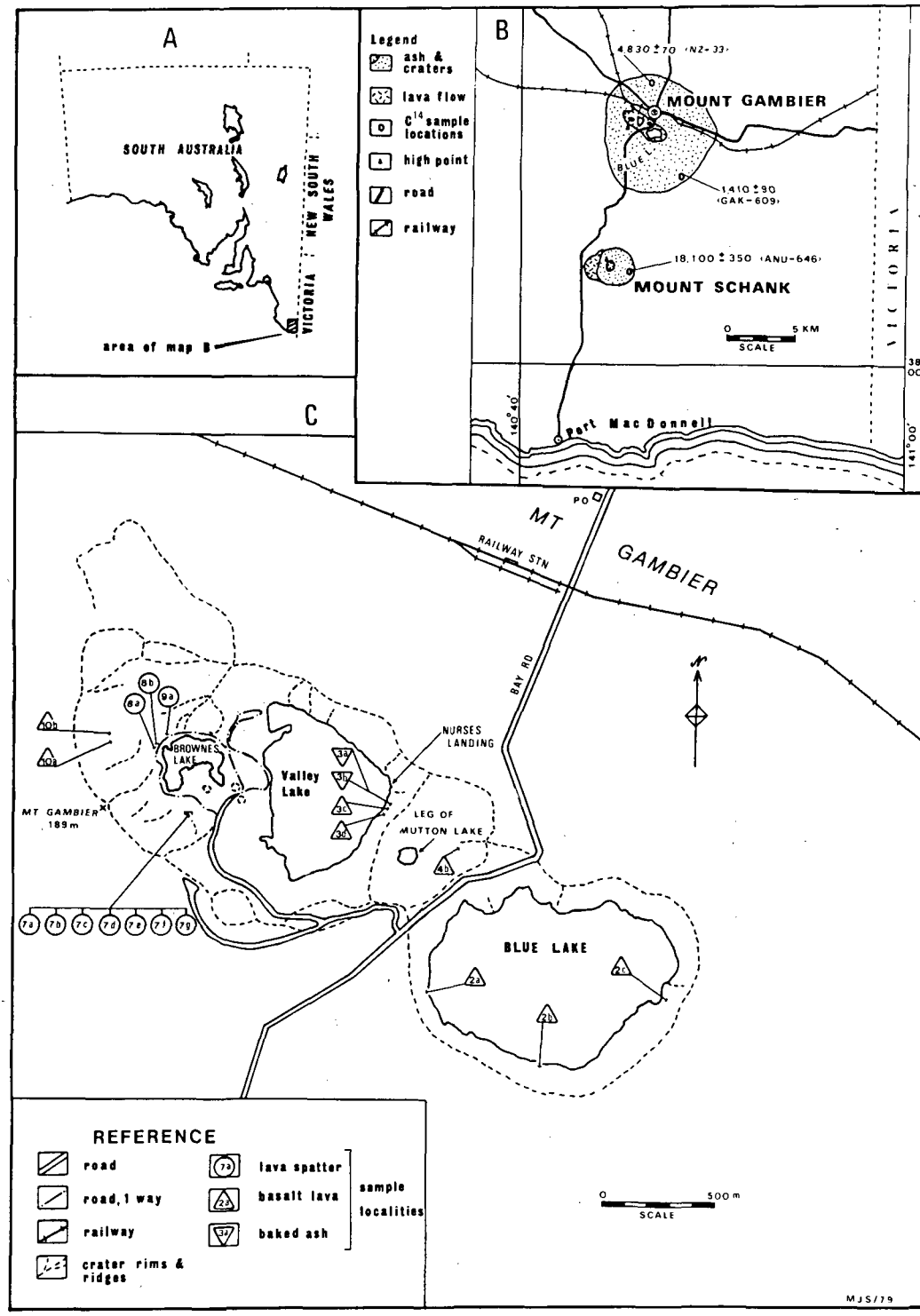


Fig. 1. (a) geographic location; (b) extent of volcanic deposits and locations of charcoal samples previously collected for radiocarbon dating; (c) details of palaeomagnetic sample localities at Mt Gambier. Samples from different rock units are distinguished by symbols; their stratigraphic relation is shown in Figure 2.

GEOLOGY

The Mount Gambier Volcanic Complex comprises a close-knit series of composite or grouped maars along a linear NW trend. The geology has been described by Sheard (1978) and may be summarised as follows. There were two separate periods of eruption, each with a distinctive style. Period one began explosively with ash and country rock ejection, and ended passively with basalt lava extrusion. Following a time break, long enough to allow complete solidification of the lava flows (at least two years), the second eruptive period commenced (Fig. 2). These eruptions were on a larger scale and more highly explosive than those of period one, due mainly to the interaction of abundant groundwater with the ascending magma column. Large open craters now containing the Blue, Valley and Brownes Lakes were the product of period two volcanism; they punctuate the ash and basalt laid down in period one. Late stage lava fountaining, restricted mainly to the Brownes Lake crater, brought activity to a close at Mt Gambier. Spatter produced by the lava fountains chilled rapidly as a glassy lava, and is preserved in forms resembling twisted rope, crusty bread, and fresh cow pats.

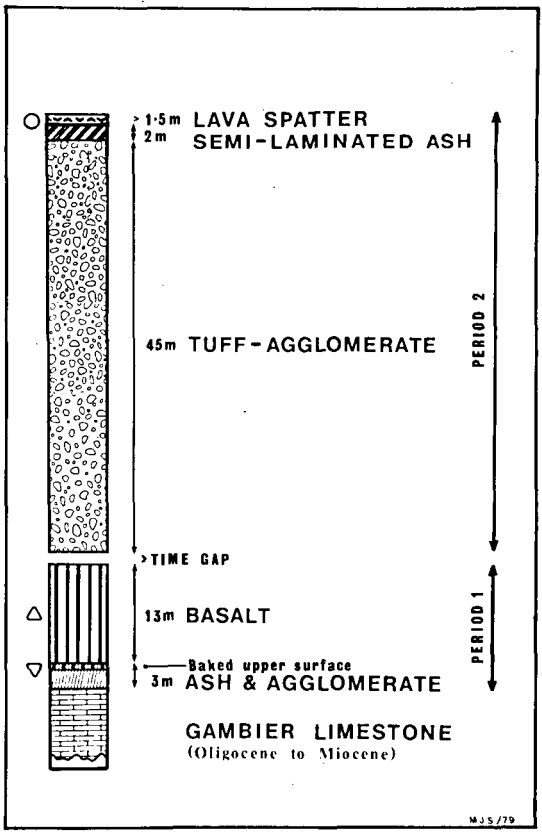


Fig. 2. Stratigraphy of the Mount Gambier Volcanic Complex, modified after Sheard (1978). Units sampled are identified by symbols to the left of the column. Sample localities are shown in Figure 1(c).

Mount Schank rises abruptly from the surrounding plain 14 km south of Mt Gambier (Fig. 1b). It is a complex maar-cone structure, where the combination of abundant groundwater and small magma source produced an edifice smaller than Mt Gambier. Initially lava flowed westward from a NW trending line or fissure, marked now by a line of small scoria cones (Fig. 3). More explosive activity followed, with the development of a larger scoria cone and small maar at the southern end of the fissure. A large hybrid maar-cone structure, straddling the small maar and almost obliterating the larger scoria cone, marks the end of activity at Mt Schank (Joyce, 1971; Sheard, 1980).

PALAEOMAGNETISM

Sampling

Oriented samples were collected from the main basalt at Mt Gambier, which is punctuated by craters and exposed at several localities, and the underlying ash baked by the basalt (Fig. 1c). The baked ash and lava would have acquired their remanent magnetisation simultaneously during cooling. Lava spatter deposits, which mark the end of period two volcanic activity at Mt Gambier and were erupted some time after the main basalt (Fig. 2), were sampled. Two embedded flint nodules were also collected, mainly for ancient field strength measurements. At Mt Schank samples were collected from the interior of the lava flow at exposed quarry faces (Fig. 3). Samples of underlying baked sand were collected for ancient field strength measurements, but they were too friable to orient.

Selected blocks 10-15 cm across were oriented before they were detached from the outcrop. A horizontal surface was first made on top of each block with plaster of paris and an aluminium levelling disc. Several lines were then inscribed in the plaster, using a magnetic compass mounted on an aluminium spacer (10 cm high). The azimuth of the lines was then measured with a sun compass of the type described by McElhinny (1973). Distortions of 10° or more were observed for magnetic compass bearings on some of the samples; all later measurements were corrected and referred to geographic north using the sun compass measurements. Cylindrical specimens (25 mm diameter, 22 mm high) for palaeomagnetic measurements were obtained by coring the blocks in the laboratory, while carefully preserving the field orientation marks.

Palaeomagnetic directions

Measurements were made with a 'Digico' slow-speed computerised magnetometer (Molyneux, 1971). Demagnetisation experiments were carried out on two specimens from each sample in a variety of alternating magnetic fields. Demagnetisation curves for Mt Gambier specimens are illustrated in Figure 4; curves for the Mt Schank specimens are similar. In many cases a substantial amount of magnetisation was removed in alternating fields less than 15 mT; soft portions such as these may change over periods of a few thousand years or even during collection, and are referred to as viscous components.

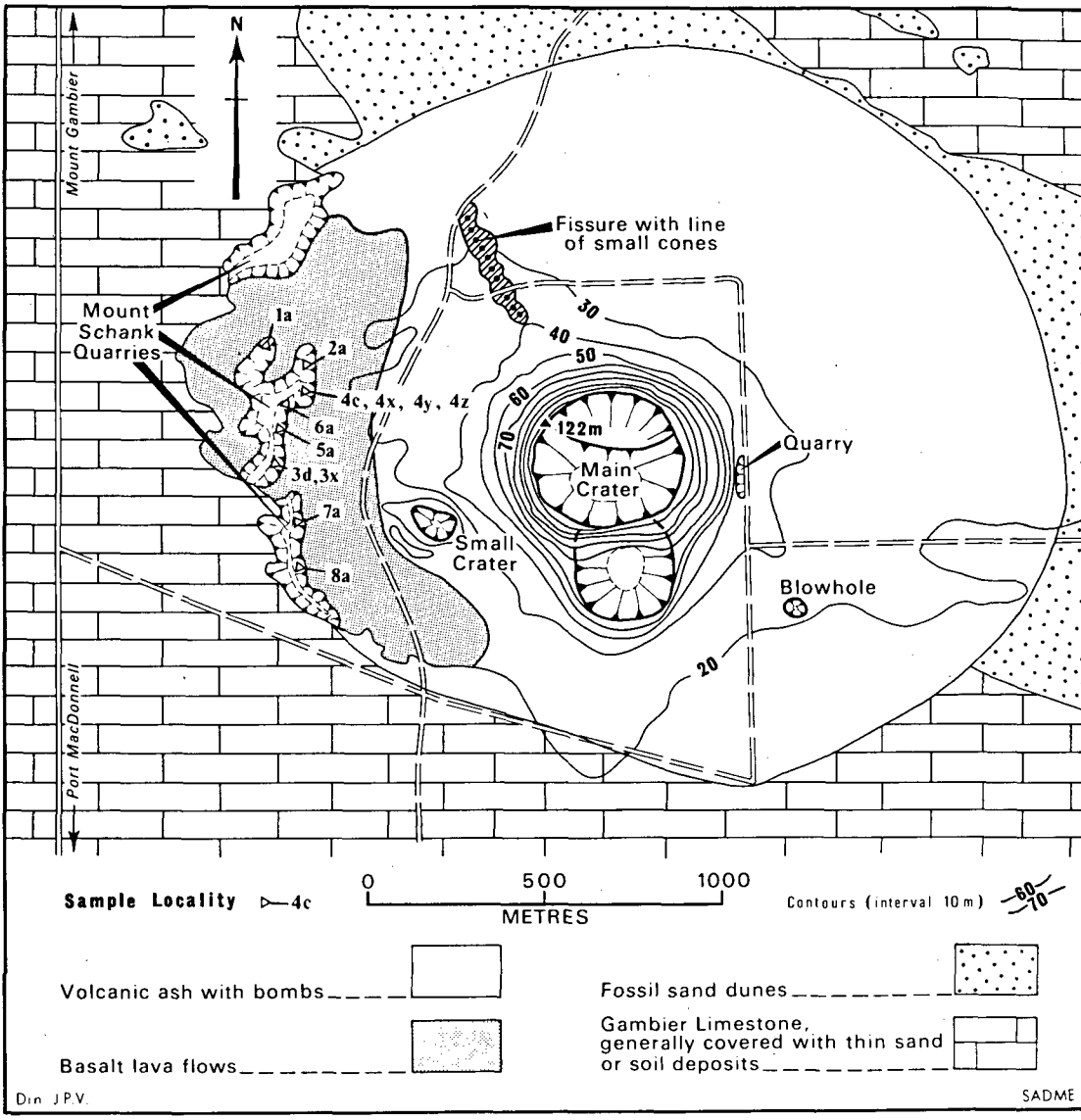


Fig. 3. Geological details and palaeomagnetic sample localities for Mt Schank. Modified after Selby & Sheard (1979).

Apart from viscous components, all specimens were found to have single main components of magnetism. Directions were calculated for each specimen by principal component analyses of measurements made during demagnetisation at peak alternating fields above 15 mT (Kirschvink, 1980). The method minimises the effects of measurement errors for each specimen. Results for the pair of specimens from each sample were then combined in the usual manner (Fisher, 1953).

Directions of magnetisation for the different samples are listed in Tables 1 and 2, and illustrated in Figure 5. Mean directions from the lava spatter, basalt and underlying baked ash at Mt Gambier are identical, despite the fact that the spatter and basalt are separated stratigraphically by a considerable thickness of tuff-agglomerate (Fig. 2). The larger degree of scatter apparent for the basalt samples is probably due to movement during the violent eruptions of period two and subsequent compaction by overlying tuff. The mean of all Mt Gambier results clearly differs from the mean direction for Mt Schank, and both directions are very different from the present-day direction (Fig. 5).

Palaeomagnetic field strengths

Samples of red-coloured baked ash from Mt Gambier and baked sand from Mt Schank were chosen for ancient field strength determinations, because they are much less likely than basalt to undergo oxidation

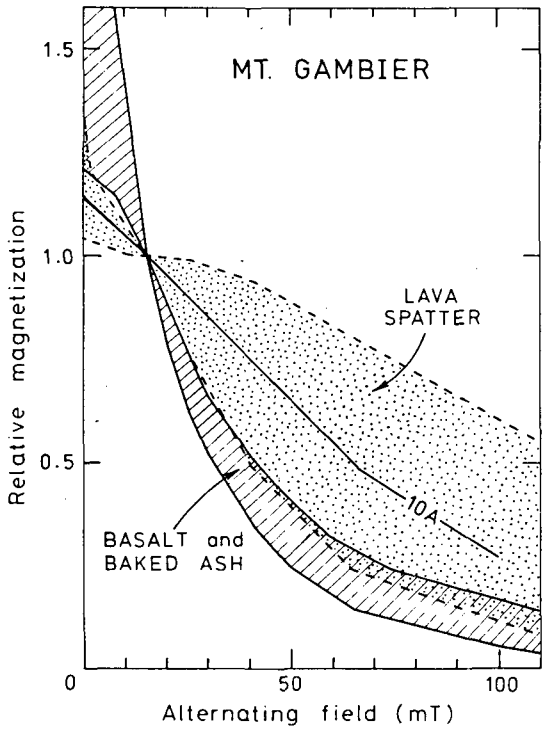


Fig. 4. Demagnetisation envelopes for Mt Gambier specimens. Demagnetisations were accomplished by tumbling the specimens in an alternating magnetic field, the peak value of which is given on the abscissa. Initial magnetisations ranged from 2 to 30 A.m^{-1}. Basalt and baked ash specimens (with the exception of those from sample 10A) have slightly lower coercivities than specimens of lava spatter; the difference may be due to the degree of oxidation.

during laboratory heating experiments. Measurements were also made on two flint nodules embedded in lava spatter at Mt Gambier. Cubic specimens (15 mm edges) were cut from the samples and subjected to a modified Thellier procedure (Thellier & Thellier, 1959; Nagata et al., 1963; Coe, 1967*a*). Laboratory heatings were carried out in a furnace at a series of progressively higher temperatures between 140°C and 680°C. Specimens were heated twice to each temperature, cooled and remeasured; the first of each pair of heatings was conducted in zero magnetic field (achieved with a surrounding coil system; McElhinny et al., 1971), and the second in the ambient laboratory magnetic field (60 μT). The first heating and cooling cycle removes a portion of the natural remanent magnetisation (NRM), and the second replaces it with a laboratory thermoremanent magnetisation (TRM). Since TRM is proportional to the magnetic field in which it is acquired, comparisons between the NRM and laboratory TRM provide estimates of ancient geomagnetic field strength.

Measurements are usually presented as NRM-TRM diagrams, in which the partial NRM remaining after heating to each temperature is plotted against the partial TRM acquired during the second heating and cooling to that temperature. In an ideal case, all the points would fall on a line with negative slope equal to the ratio between NRM and TRM, which would be the same as the ratio between the ancient and laboratory magnetic fields (Nagata et al., 1963; Coe, 1967*a*). In practice, however, the points are scattered by experimental error, weathering, changes in the magnetic minerals during the laboratory heatings, and other forms of non-ideal behaviour (Coe, 1967*b*). Therefore, lines are fitted to those points which appear to lie on a straight line; the NRM is generally excluded because of the possible presence of viscous components, and high-temperature points that deviate are also excluded. We performed trial analyses on all the NRM-TRM diagrams, using the 'maximum likelihood' method of Kendall & Stuart (1973), excluding the NRM always and then progressively excluding the higher-temperature points until the width of the calculated confidence limits approached a minimum.

A selection of the NRM-TRM diagrams with finally-fitted lines is shown in Figure 6, and all the results are listed in Table 3. The flint specimens exhibit curved NRM-TRM diagrams with no apparent linear segments; this is probably an intrinsic property of the material, and may be due to the presence of multi-domain grains of magnetite (Levi, 1977). Their magnetisation directions and characteristics do, however, show that the flint nodules cooled from above 600°C in the positions in which they were found. NRM-TRM diagrams for the Mt Gambier baked ash generally show a decrease in partial TRM above about 550°C, which suggests that magnetite may have been oxidised to hematite (which is less magnetic) during heatings to those temperatures. Specimens of baked sand from Mt Schank generally show non-linear behaviour, with strongly increasing partial TRM, above about 450°C (e.g. specimen 4X11 in Fig. 6.). This is attributed to the presence of iron hydroxides (such as goethite) which would be converted to hematite during laboratory heating (Barbetti et al., 1977). The Mt Schank specimens also show decreases in partial TRM above about 550°C. The fact that results between specimens become more consistent after deviant high-temperature points are excluded, here and in other documented examples (Barbetti & Flude, 1979*a*, 1979*b*), is an argument for the validity of the procedure used.

Mean values for ancient strengths are given in Table 3, along with 95% confidence limits (standard errors multiplied by appropriate values of Student's *t*). Estimates are also given for the strength of hypothetical sources at the Earth's centre which would produce the ancient field strengths. Virtual axial dipole moments are calculated on the assumption that the dipole axis coincides with the Earth's geographic axis, and virtual dipole moments on the assumption that the dipole axis is tilted so as to give the measured palaeomagnetic inclination at the site (Barton et al., 1979). Values may be compared with the present-day dipole moment of 8 x 10^{22} Am^2.

DISCUSSION

Perhaps the most interesting observations to emerge from our study are the identical nature of

TABLE 1. Palaeomagnetic directions for Mt Gambier after cleaning.

Sample	Declination (°)	Inclination (°)	Pole (°E)	Position (°N)
MT. GAMBIER				
Lava Spatter (O):				
7B	351	-56		
7D	357	-55		
7E	1	-58		
7F	357	-56		
7G	2	-57		
8A	357	-59		
8B	358	-55		
9A	3	-59		
Site Mean :	358	-57	61	88½
(N=8, R=7.9926, k=947, α_{95}=1.8°)			(dp=1.9°, dm=2.6°)	
Basalt (Δ) and underlying baked contact (∇):				
2A Δ	0	-57		
2B Δ	1	-47		
2C Δ	348	-61		
3A ∇	358	-58		
3B ∇	358	-57		
3C Δ	357	-61		
3D Δ	4	-54		
4B Δ	354	-62		
10AΔ	2	-59		
10BΔ	341	-58		
Site Mean :	357	-57½	47	87½
(N=10, R=9.9553, k=201, α_{95}=3.4°)			(dp=3.6°, dm=5.0°)	
Mean of all Mt. Gambier results:	357	-57	52	88
(N=18, R=17.9474, k=323, α_{95}=1.9°)			(dp=2.0°, dm=2.8°)	

Mean declinations (east from geographic north) and inclinations (from the horizontal, negative sign indicating vectors above the horizontal) are listed for each sample. Site (rock unit) mean values are given with Fisher (1953) statistical parameters; N is the number of samples, R is the resultant of N unit vectors, k is the precision parameter and α_{95} the radius of the circle of 95% confidence for the true value. Corresponding Virtual geomagnetic pole positions are also listed; dp and dm are the uncertainties along and perpendicular to the great circle between the pole and the site.

TABLE 2. Palaeomagnetic directions for Mt Schank after cleaning.

Sample	Declination (°)	Inclination (°)	Pole (°E)	Position (°N)
MT. SCHANK				
Basalt (□) :				
1A	8	-53		
2A	4	-54		
3D	9	-55		
4C	9	-56		
5A	13	-57		
6A	8	-56		
7A	8	-54		
8A	15	-56		
Site Mean :	9	-55	216	82
(N=8, R=7.9939, k=1150, α_{95}=1.6°)			(dp=1.6°, dm=2.3°)	

Notation as in Table 1.

palaeomagnetic directions from the basalt and lava spatter at Mt Gambier and the difference between those and the direction for Mt Schank. The first observation suggests that the time gap between periods one and two at Mt Gambier was fairly short (possibly less than a century) and the second firmly indicates that Mt Schank did not erupt at the same time as Mt Gambier. If we assume a reasonable maximum rate for past declination changes in southeastern Australia, say 3° per century, our results indicate that ages for Mt Gambier and Mt Schank should differ by three centuries or more.

Comparisons may also be made between palaeomagnetic results for Mt Gambier and Mt Schank and other late Pleistocene and Holocene results from southeastern Australia (Barbetti, 1977; Barton, 1978; Barbetti, 1980). As noted earlier (Fig. 5), palaeomagnetic directions for Mt Gambier and Mt Schank are very different from the present-day direction; they also differ significantly from other palaeomagnetic directions for the last 1000 years. Palaeomagnetic field strengths for Mts Gambier and Schank are higher than the present-day value, and very different from the low field strengths observed for the period 7000 to 5000 years ago. Possible ages for Mts Gambier and Schank therefore are restricted to the very late Pleistocene, early Holocene or the period 5000-1000 years ago.

ACKNOWLEDGMENTS

Field sample collection from the Blue Lake crater wall was made possible by Mr Don Ide and the staff of the Engineering & Water Supply Department, and sampling at Mt Schank was done with the cheerful cooperation of Mr Sam Sutton, site manager for Mt Schanck Quarries Pty. Ltd. Mr B. Smith assisted with sample collection, and Dr F. Chamalaun kindly allowed us to core the samples at Flinders University. We thank Dr M. W. McElhinny for allowing us to carry out measurements at the Palaeomagnetic Laboratory of the Australian National University, and Dr C. T. Klootwijk for help with data processing. This work was carried out while M.B. was a Research Fellow at the Physics Department, University of Adelaide; encouragement from Professor J. R. Prescott and Dr D. J. Sutton and support from the Australian Research Grants Committee are gratefully acknowledged. We also thank Mrs J. Noel for typing the manuscript. This paper is published with the permission of the Director General, South Australian Department of Mines and Energy.

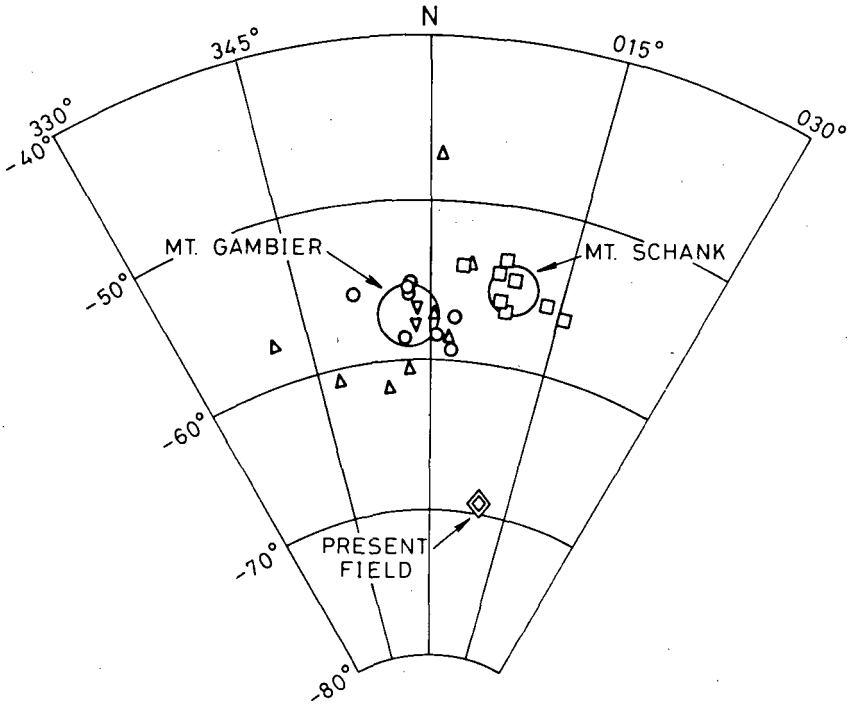


Fig. 5. Stereographic projection of sample directions of magnetisation; symbols are those used in Figures 1, 2 and 3. Declinations are relative to geographic north and inclinations are above the horizontal. Circles of 95% confidence for the true values (Tables 1 and 2), and the present-day direction at the locality, are also shown.

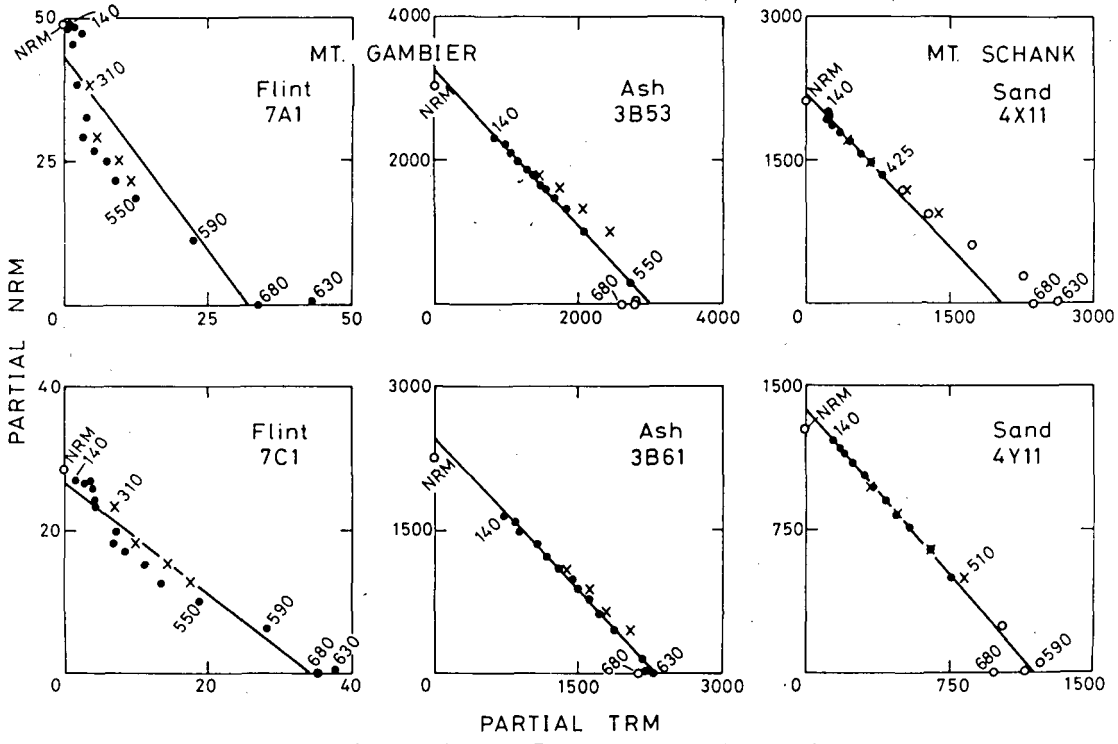


Fig. 6. Representative NRM-TRM diagrams for Mt Gambier and Mt Schank. Units are $\mu A.m^2.kg^{-1}$ and numbers near the points are temperatures in °C. The NRM and measurements made in the usual manner at progressively increasing temperatures are represented by circles, whereas crosses mark repeat TRM determinations made after heatings had progressed to 590°C. Lines have been fitted to points shown as solid circles, using the 'maximum likelihood' method of Kendall & Stuart (1973).

TABLE 3. Summary of palaeomagnetic field strength measurements.

Specimen	Temperature range (°C) (n points)	Ancient field strength (μT)	Dipole Moment (10^{22} A.m^2) Virtual Axial	Virtual
MT. GAMBIER				
Flint inclusions in lava spatter:				
7A1	140-680 (15)	∿ 81		
7C1	"	∿ 46		
7C2	"	∿ 52		
7C3	"	∿ 46		
Baked ash under basalt:				
3B51	140-630 (14)	64±2		
3B53	140-550 (12)	65±2		
3B6	140-550 (12)	66±5		
3B61	140-630 (14)	64±3		
3B64	140-550 (12)	66±3		
	Mean	65±1	11.5±0.2	11.5±0.4
MT. SCHANK				
Baked sand under basalt:				
3X2	140-425 (9)	59±6		
3X4	140-425 (9)	62±6		
4X11	140-425 (9)	64±8		
4X2	140-425 (9)	62±8		
4Y11	140-510 (11)	70±1		
4Y22	140-390 (8)	74±6		
4Z2	140-425 (9)	63±6		
	Mean	65±5	11.5±0.9	11.8±0.9

Initial numbers and letters (e.g. 7A) identify the sample and subsequent numbers the position and depth below the top of the sample (e.g., specimens 1 and 2 from side by side on top and specimens 11,12 etc. successively below 1). Column 2 gives the temperature range and number of points used in the final line-fitting. Ancient field strengths and means are given with 95% confidence limits. Corresponding Virtual Axial and Virtual Dipole Moments are also listed; 95% confidence limits for the former correspond to the uncertainty in field strength alone, but those for the latter include contributions from uncertainties in both field strength and palaeomagnetic inclination.

REFERENCES

BARBETTI, M., 1977: Measurements of Recent geomagnetic secular variation in southeastern Australia and the question of dipole wobble. *Earth planet. Sci. Lett., 36,* 207-18.

BARBETTI, M., 1980: Geomagnetic strength over the last 50 000 years and changes in atmospheric ^{14}C concentration: emerging trends. *Radiocarbon, 22,* 192-9.

BARBETTI, M. & FLUDE, K., 1979*a*: Palaeomagnetic field strengths from sediments baked by lava flows of the Chaîne des Puys, France. *Nature, 278,* 153-6.

BARBETTI, M. & FLUDE, K., 1979*b*: Geomagnetic variation during the late Pleistocene period and changes in the radiocarbon time-scale. *Nature, 279,* 202-5.

BARBETTI, M. F., MCELHINNY, M. W., EDWARDS, D. J. & SCHMIDT, P. W., 1977: Weathering processes in baked sediments and their effects on archaeomagnetic field strength measurements. *Phys. Earth planet. Interiors, 13,* 346-54.

BARTON, C. E., 1978: *Magnetic studies of some Australian lake sediments.* Ph.D. thesis, Australian National University, Canberra (*unpubl.*).

BARTON, C. E., MERRILL, R. T. & BARBETTI, M., 1979: Intensity of the Earth's magnetic field over the last 10 000 years. *Phys. Earth planet. Interiors, 20,* 96-111.

COE, R. S., 1967*a*: Paleo-intensities of the Earth's magnetic field determined from Tertiary and Quaternary rocks. *J. Geophys. Res., 72,* 3247-62.

COE, R. S., 1967*b*: The determination of paleo-intensities of the Earth's magnetic field with emphasis on mechanisms which could cause non-ideal behavior in Thellier's method. *J. Geomag. Geoelectr., 19,* 157-79.

FERGUSSON, G. J. & RAFTER, T. A., 1957: New Zealand ^{14}C age determinations—3. *N.Z. J. Sci. Tech., 38B,* 732-49.

FISHER, R. A., 1953: Dispersion on a sphere. *Proc. R. Soc. London, A217,* 295-305.

JOYCE, E. B., 1971: A study of maars in the post-Miocene volcanics of southeastern South Australia. *Abstracts 43rd ANZAAS Congress, Brisbane, Qld.,* p. 72.

KENDALL, M. G. & STUART, A., 1973: *The advanced theory of statistics* (3rd. ed.), vol 2, Griffin, London.

KIGOSHI, K. & KOBAYASHI, H., 1966: Gakushuin natural radiocarbon measurements V. *Radiocarbon, 8,* 54-73.

KIRSCHVINK, J. L., 1980: The least-squares line and plane and the analysis of palaeomagnetic data. *Geophys. J. R. astron. Soc., 62,* 699-718.

LEVI, S., 1977: The effect of magnetite particle size on paleointensity deterfinations of the geomagnetic field. *Phys. Earth planet. Inter., 13,* 245-59.

MCELHINNY, M. W., 1973: *Palaeomagnetism and plate tectonics,* University Press, Cambridge.

MCELHINNY, M. W., LUCK, G. R. & EDWARDS, D., 1971: A large volume magnetic field free space for thermal demagnetization and other experiments in palaeomagnetism. *Pure Appl. Geophys., 90,* 126-30.

MOLYNEUX, L., 1971: A complete result magnetometer for measuring the remanent magnetization of rocks. *Geophys. J. R. astron. Soc., 24,* 429-33.

NAGATA, T., ARAI, Y. & MOMOSE, K., 1963: Secular variation of the total geomagnetic force during the last 5000 years. *J. Geophys. Res., 68,* 5277-81.

POLACH, H. A., HEAD, M. J. & GOWER, J. D., 1978: ANU radiocarbon date list VI. *Radiocarbon, 20,* 360-85.

SELBY, J. & SHEARD, M. J., 1979: Volcanoes of the Mount Gambier area. *Mineral Information Series (Pamphlet),* Geol. Surv. S. Aust., 12p.

SHEARD, M. J., 1978: Geological history of the Mt Gambier volcanic complex, southeast South Australia. *Trans. R. Soc. S. Aust.,* 102, 125-39.

SHEARD, M. J., 1980: Quaternary volcanism in south-eastern South Australia. Rept. Bk. No. 80/20 geol. Surv. S. Aust. (unpubl).

SOLOMON, M., 1951: The volcanic deposits of the south-east of South Australia as sources of quarry stone. *Min. Rev., 93,* 133-7.

SPRIGG, R. C., 1952: The geology of the south-east province with special reference to Quaternary coast line migrations and modern beach developments. *Bull. geol. Surv. S. Aust., 29,* 30-62.

THELLIER, E. & THELLIER. O., 1959: Sur l'intensité du champ magnétique terrestre dans le passé historique et géologique. *Annales de Géophysique, 15,* 285-376.